Comparison of the Learning Curve and Adaptive Behavior from Kids to Adults who Create Mobile-Apps and Little Robots Using Block-Programming

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Motivation

- **Problem:** Peruvian schools keep technology out of the classroom and don't teach programming at early ages. Young people first learn it at university.
- **The gap:** Kids of 8 already use tablets and phones daily, but for games, social networks and videos, not for learning.
- **Our approach:** Free courses, *"Robotics for everyone"* and *"Game development for everyone"*, that hide computational thinking and algorithm design inside block programming.

Kids	Teens	Juniors	Adults
6–9	10–12	13–17	18+

First programming at university: age 20–22

How much do learning speed and retention differ across ages under the same learning conditions?

Study Design

≈400	25	16	57.2%
students, about 100 per age group	people in each class group	lessons of 3 hours a day	of Peru's population is aged 9–45

Every group had the **same curriculum, materials, teachers and lessons**, covering C and Python programs and Arduino robots.

Learning Curve
How learning increases with experience: do students solve problems better than before?

Adaptive Behavior
How students accept or reject new concepts, and start using them in their own examples and jokes.

Methodology

Four techniques, used in sequence

Script language — Everyday questions with a concept hidden in the answers ▸
Metaphors — Common examples that explain computing concepts ▸
Mind maps — Brainstormed ideas connected like a graph ▸
Block programming — Code joined as blocks, then the jump to real code

"What is an algorithm?" asked by age

KIDS
"If you want to go to the bathroom, what should you say? How do you clean your desk?"

TWEENS
"Have you ever played a game on your computer? Have you heard about PSP or XBOX? Do you know what Mario Kart is?"

TEENS
"Do you know how to solve a linear equation? Which rules do you follow when you play football? Have you ever played guitar?"

ADULTS
"Have you ever developed something? Have you heard about programming languages? Do you know what software means?"

The questions look different, but in the background they are very similar: every group answered "Yes!" and explained a sequence of steps.

Novice	Advanced beginner	Competence	Proficiency	Expert

Five stages of programming skills: all our students started as novices

Building Mini Robots

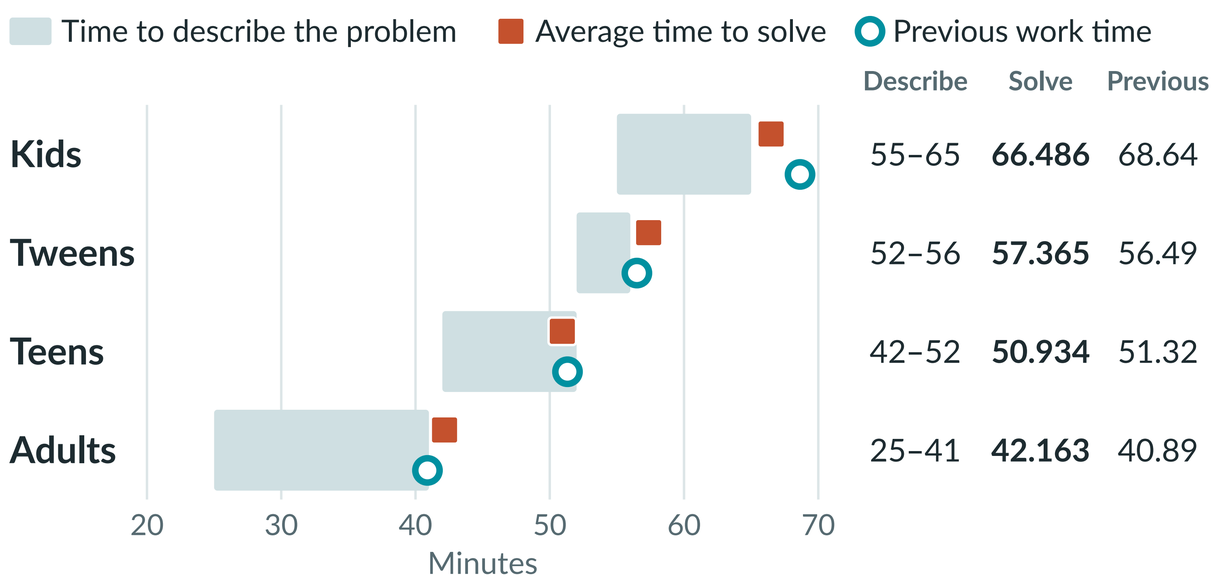
Otto
programmed on **BitBloq**

Kyo
programmed on **Tinkercad**

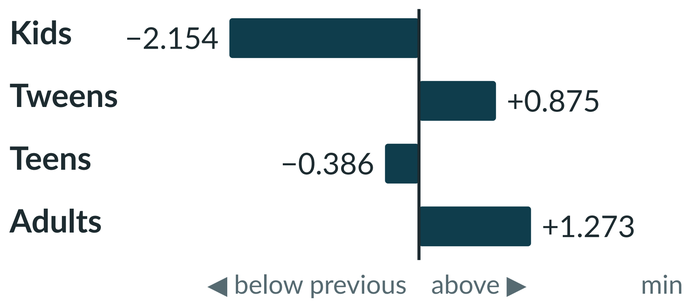
- Students simulate circuits online and program movements and sensors with **blocks and C**.
- They then translate their **mind maps into a program**, in blocks or a programming language.
- **Programming speed improves** once students understand the logic of the mind map they drew.

Results

How long does it take to learn a new concept with examples?



Solve time minus previous work



24.3 min
less for adults than for kids to solve. Times stay within about **3%** of previous work, and fall below it for kids and teens.

Approximating the average time
A learning-curve formula is compared with the real average. **K**: hours to understand the first unit, from the script-language part. **x = 4**: the 4th unit. **b**: percent of learning, from the literature.

Main Findings

- **Tweens and teens** have more ability to understand new concepts using **games as metaphors**.
- **Adults** have a strong learning speed, based on **past experience**.
- **Kids to juniors** learn fast, but **forget concepts** in a short period of time.

Learning time doesn't depend on a specific topic: it depends on the methodology applied to teach it.

Acknowledgments & References

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